

Case Study: Why Two ILDW/OPT Pairs Look the Same and One Looks Different

Summary

The key distinction across the three case-study patches is the size of the pixel-wise change between the ILDW field and the optimized field

$$|R_{\text{OPT}}(x, y) - R_{\text{ILDW}}(x, y)|.$$

For the first two patches, the optimizer starts from ILDW and only makes very small adjustments. As a result, the rainy error maps for ILDW and OPT look almost identical. For the third patch, the optimizer moves much farther away from the ILDW initialization, and that larger change is also accompanied by a clear improvement against ground truth. Consequently, the ILDW and OPT rainy maps look visibly different.

Interpretation

- **Patch 1:** RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301280900_patch000 looks the same because the optimizer changes the field only by a few thousandths to a few hundredths of mm/h for most pixels.
- **Patch 2:** RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301310900_patch000 follows the same pattern. The optimizer improves the objective slightly, but the actual map remains very close to ILDW.
- **Patch 3:** RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301311600_patch000 is different because the optimizer makes a much larger field update over a large fraction of the pixels. That is why the rainy maps visibly diverge.

Quantitative Justification

Statistic	202301280900 patch000	202301310900 patch000	202301311600 patch000
Median $ OPT - ILDW $ (mm/h)	0.00076	0.00182	0.20513
$p_{90}(OPT - ILDW)$ (mm/h)	0.00764	0.01445	0.61398
$p_{95}(OPT - ILDW)$ (mm/h)	0.01631	0.03352	0.76036
$p_{99}(OPT - ILDW)$ (mm/h)	0.07920	0.12804	1.19901

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Statistic	202301280900 patch000	202301310900 patch000	202301311600 patch000
Mean $ OPT - ILDW $ (mm/h)	0.00483	0.00820	0.28199
Rainy mean $ OPT - ILDW $ (mm/h)	0.00661	0.00867	0.30534
Non-rainy mean $ OPT - ILDW $ (mm/h)	0.00179	0.00559	0.12701
Correlation between ILDW and OPT fields	0.999935	0.999707	0.961674
Pixels with $ OPT - ILDW > 0.1$	0.73%	1.46%	71.99%
Pixels with $ OPT - ILDW > 0.5$	0.03%	0.03%	18.11%
Pixels with $ OPT - ILDW > 1.0$	0.002%	0.002%	1.92%
ILDW rainy MAE (mm/h)	0.80682	0.56389	0.35919
OPT rainy MAE (mm/h)	0.80493	0.56113	0.32423
Rainy MAE ratio OPT / ILDW	0.99765	0.99512	0.90266
ILDW all-pixel MAE (mm/h)	0.57718	0.57696	0.34235
OPT all-pixel MAE (mm/h)	0.57588	0.57434	0.30110
All-pixel MAE ratio OPT / ILDW	0.99774	0.99546	0.87950
Optimizer iterations used	208	19	300

Reading the Table

The first two cases are visually similar because their median and upper-quantile changes are tiny: even the 95th percentile of $|OPT - ILDW|$ stays below 0.04 mm/h. In contrast, for the third case the median change is already 0.205 mm/h and the 90th percentile exceeds 0.6 mm/h. This means a large portion of the field is being reshaped, which is exactly why the rainy maps look different to the eye.

The MAE ratios tell the same story. In the first two patches, OPT improves rainy MAE only marginally, to about 99.8% and 99.5% of the ILDW value¹, while in the third patch OPT reduces rainy MAE to about 90.3% of the ILDW value, which is a much more substantial gain.

¹Equivalently, this is a relative rainy-pixel MAE improvement of about 0.2% to 0.5%. Here, $MAE = \text{mean}(|\text{prediction} - \text{GT}|)$. For 202301280900_patch000, rainy-pixel MAE decreases from 0.80682 to 0.80493, i.e. $(0.80682 - 0.80493)/0.80682 \approx 0.23\%$. For 202301310900_patch000, rainy-pixel MAE decreases from 0.56389 to 0.56113, i.e. about 0.49%.

Paths Within the norm Run

Patch 1: RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301280900_patch000

- j_behavior: report/images/j_behavior/opt_norm_ildw_mult_ildw_init/RAD_OPERA_HOURLY_RAINFALL_
- ILDW rainy map: report/images/patch_error_maps/ILDW/RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_
- OPT rainy map: report/images/patch_error_maps/OPT_NORM_ILDW_MULT_ILDW_INIT/RAD_OPERA_HOURLY_

Patch 2: RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301310900_patch000

- j_behavior: report/images/j_behavior/opt_norm_ildw_mult_ildw_init/RAD_OPERA_HOURLY_RAINFALL_
- ILDW rainy map: report/images/patch_error_maps/ILDW/RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_
- OPT rainy map: report/images/patch_error_maps/OPT_NORM_ILDW_MULT_ILDW_INIT/RAD_OPERA_HOURLY_

Patch 3: RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301311600_patch000

- j_behavior: report/images/j_behavior/opt_norm_ildw_mult_ildw_init/RAD_OPERA_HOURLY_RAINFALL_
- ILDW rainy map: report/images/patch_error_maps/ILDW/RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_
- OPT rainy map: report/images/patch_error_maps/OPT_NORM_ILDW_MULT_ILDW_INIT/RAD_OPERA_HOURLY_